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GenAI Based Framework for Personalized Training Recommender in Energy Sector

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Abstract

As the energy industry rapidly evolves, the challenge of keeping employees up to date is growing. Traditional training plans and programs no longer meet the demands of modern roles which led to outdated, repetitive, and generic learning outcomes. This paper investigates the effectiveness of Generative AI based training recommender systems aiming to provide tailored, personalized course recommendations to yield higher employee engagement, and faster upskilling. Natural language processing, machine learning, and knowledge graph techniques are integrated to better comprehend employees' current skills and needs. By using a two-pass filter, the system first analyzes the profile of the employee, then delivers customized training material suggestions, while Large Language Model (LLM) fine-tuning ensures that recommendations remain up-to-date and contextually accurate. The research findings are promising, as the employees who received training using the system demonstrated more engagement and better job performance than those using traditional methods. The effectiveness of the Generative AI-based training recommender system was evaluated through a survey of more than two hundred employees, comparing the perceived usefulness, relevance, and impact of the recommended courses on their job performance and engagement between the new system and traditional training methods. The results were analyzed using both qualitative and quantitative methods, including statistical comparisons of mean scores and standard deviations, to confirm the significant benefits of the new system.

Introduction

In today's rapidly evolving digital landscape, employee development has become more critical than ever. From the high-stakes decisions to the safety of field operations, the energy sector relies on a workforce that is constantly evolving in its skillset. Traditional training programs, however, often fall short in meeting the needs of these dynamic sectors[1]. Relying on general and static training material has proven to be insufficient in preparing professionals for the latest emerging technologies, which leads to disengaged learners and compromised readiness in critical job functions [2]. Bridging this gap, our research proposes a novel, multi-disciplinary training recommender system specifically crafted for the energy sector. The proposed framework integrates Natural Language Processing (NLP), Machine Learning

(ML), and knowledge graphs enhanced with LLMs to enable personalized employee development. These learning journeys are built from rich metadata rather than solely surface-level metadata such as job title and years of experience. By training this system on historical data, skill gaps and knowledge deficiencies are identified through machine learning algorithms to provide relevant training recommendations and foster a learning environment where training is not perceived as an obligation, but rather an engaging and fruitful experience. A two-pass filter architecture is employed where the first pass analyzes employee profiles for the extraction of prior training, role context, and identified competency gaps. The second pass is for the semantic matching of content using NLP and graph reasoning to ensure contextual relevance. These two steps ensure recommendations are not only relevant but also contextually relevant, matching the exact skill set and operational needs of the employee. We anticipate several outcomes: higher employee engagement, faster acquisition of capabilities and better on-the-job performance. More importantly, we view this as allowing each employee, regardless of role or location, to receive the training they need, when needed. As the energy sector embarks on the digital transformation wave, the convergence of generative AI with workforce development is not just a possibility but a necessity. Ultimately, the proposed solution not only enriches learning experiences and supports succession planning, but it also encourages an environment with proactive growth where development is not imposed but strategic, personalized, and efficient.

Overview on the paper structure

The paper is organized as follows. Next section provides a literature review of some relevant research papers that delves into various debates and key concepts related to predicting drilling equipment's failure time and real-time anomaly detection in drilling data. The following section describes the methodology that outlines the experiment plan, dataset description, and observations. The experiment results section analyses the result achieved in the experiments and findings. Lastly the paper concludes by highlighting the experiment findings and future direction.

Research Problem Definition

The energy industry is facing a significant challenge in providing personalized and effective training to its workforce, resulting in suboptimal skill acquisition and performance. The rapid evolution of the industry necessitates advanced training methods that can keep pace with the changing needs of the workforce. However, traditional training methods have limited ability to provide personalized recommendations, leading to inefficient use of time and resources. The unique challenges posed by the energy industry, such as the need for context-specific training and the complexity of employee roles and responsibilities, require specialized approaches to improve training effectiveness.

The lack of progress in advancing training technologies specific to the energy industry has resulted in limited adoption of innovative solutions, leading to increased training costs and decreased employee engagement. Traditional methods have struggled to handle the complexities of employee data, such as diverse skill sets and learning preferences, effectively. Therefore, there is a pressing need for novel approaches that leverage advanced Generative AI (GenAI) and machine learning techniques to enhance training personalization and effectiveness in the energy industry.

Research Objectives

The primary objectives of this research are to:

1. **Develop a GenAI-Based Training Recommender System:** Design and implement a GenAI-based training recommender system incorporating natural language processing, machine learning, and knowledge graph techniques tailored to the unique challenges posed by the energy industry.

2. **Predict Personalized Training Recommendations:** Utilize the developed system to accurately predict personalized training recommendations for employees based on their skill gaps, roles, and responsibilities.
3. **Detect Skill Gaps and Knowledge Deficiencies:** Implement machine learning and GenAI techniques to identify skill gaps and knowledge deficiencies in employees, including areas where additional training is required.
4. **Evaluate System Performance:** Evaluate the effectiveness of the Generative AI-based training recommender system in providing personalized and relevant course recommendations to employees, including:
 - Assessing the perceived usefulness, relevance, and impact of the recommended courses on employees' job performance and engagement.
 - Comparing the performance of the new Generative AI-based system with the traditional training method.
 - Measuring the level of engagement and satisfaction of employees with the recommended courses.
 - Determining the likelihood of employees recommending the training program to their colleagues.

Literature Review

Deep Learning Approaches for Course Recommendation Systems in Employee Training

The integration of deep learning into personalized course recommendation systems has garnered increasing attention, particularly in the context of professional development and lifelong learning. Traditional collaborative filtering and content-based approaches often fall short in capturing complex, non-linear user-item relationships. To address these limitations, several recent studies have proposed deep neural network (DNN)-based frameworks tailored to educational environments, including systems that can be adapted to corporate training.

Ren et al. (2022) present a robust LSTM-Attention-based multimodal course recommendation framework designed to capture both temporal and semantic features from multiple data modalities, such as course descriptions and learner histories. Their model incorporates long short-term memory (LSTM) layers to retain user behavior sequences and attention mechanisms to weigh relevant features dynamically. This architecture allows for the nuanced modeling of learner interests over time and is especially effective in settings where learner behavior exhibits sequential dependencies, such as onboarding or skill progression tracks in corporate environments. Their work demonstrates significant performance gains over baseline models using real-world educational datasets [1].

In a more targeted application of collaborative filtering, Mulyana and Rumaisa (2020) utilize Neural Collaborative Filtering (NCF) to recommend courses based on user-course interaction matrices. Unlike traditional matrix factorization, NCF uses a deep feed-forward neural network to learn complex interaction functions between user and course embeddings. Their system was implemented within an online learning environment, highlighting its potential for corporate learning platforms seeking scalable and real-time personalization. This study emphasizes the flexibility and adaptability of neural approaches in handling sparse data and cold-start problems commonly encountered in employee training contexts [2].

Zhang et al. (2022) propose a personalized MOOC recommendation framework combining big data technologies and deep learning techniques, specifically using BERT for semantic content understanding. Their system processes unstructured course content and learner profiles to match users with highly relevant courses. This model not only addresses semantic mismatches in traditional keyword-based systems but also leverages distributed computing to enable real-time scalability—an essential feature for enterprise-

grade applications. The use of BERT allows for context-aware understanding of course materials, enhancing recommendation accuracy in domains where terminology and content vary significantly, such as technical and soft skill training [3].

The proposed systems in the two papers address these limitations by incorporating organizational preferences, user profiles, and domain expertise. The use of NLP and text analytics enables the analysis of large amounts of data, including training history and user feedback. The ensemble machine learning approach used in the second paper combines multiple models to provide more accurate recommendations. The incorporation of subject matter expert knowledge and user profile analysis ensures that the recommendations are relevant and effective [4][5].

These studies collectively illustrate the evolution of course recommendation systems from basic collaborative filtering toward intelligent, multimodal deep learning architectures. While Ren et al. emphasize the temporal dimension and sequence modeling, Mulyana and Rumaisa focus on interaction modeling through neural collaborative filtering, and Zhang et al. integrate deep semantic understanding and scalable system architecture. Together, they provide a strong foundation for developing advanced course recommendation systems in employee training ecosystems.

Generative AI in Training and Education

The growing complexity of industrial environments, particularly in the energy sector, demands smart and adaptive training solutions. The recent advances in Generative AI (GenAI) and Large Language Models (LLMs) have created new opportunities for improving personalization and effectiveness of workforce training. The following section provides an overview of several key AI-powered training recommenders studies analyzing their strategies, outcomes, and transferability to the suggested GenAI-based system.

Li et al. [6] introduced a multi-view deep learning architecture for corporate personalized training recommendation. It integrates employee profile, learning history, and course information by two sub-networks via neural networks. One captures static user features (role, department, tenure), and the other processes dynamic interactions (course taken, ratings). A joint embedding layer integrates them for trusted prediction. Their experiments, on a corporate dataset from a global energy company, showed that their method improved the F1-score by 18% compared to collaborative filtering algorithms. Perhaps most significantly, the system helped bridge personal skills gaps faster, a factor to the goal of real-time adaptive learning for high-risk industries like oil and gas.

Zhang et al. [7] developed a domain-specific LLM fine-tuning pipeline to push forward training recommendations in one other study. The work focuses on fine-tuning transformer-based LLMs with internal documentation, competency models, and job profiles of an engineering company. The authors demonstrate that fine-tuning improves contextual comprehension by enabling the model to generate role-specialized training flows, even for mixed job positions. Their recommender system uses semantic search and prompt engineering techniques to match courses to learner intent. Field trials for three departments found 23% increase in training attendance and 19% reduction in drop-outs. The study highlights the importance of in-house LLM tuning to ensure context correctness, being perfectly in accordance with your system design requirements.

Chatterjee et al. [8] developed a two-pass filter system to optimize training recommendations in business learning systems. In the first pass, rule-based filtering is done based on the role, geolocation, and skill gap analysis, and in the second pass, a BERT architecture is utilized to rank entities based on semantic similarity. The approach significantly reduced noisy training recommendations (by 35%) as well as boosted learner satisfaction scores. Their solution also incorporated feedback loops to iteratively enhance skill profiles, allowing engineers to modify the recommendations as needs in employees evolved. The model was deployed in a multinational oilfield services company, demonstrating scalability and responsiveness of the model within large, dynamic environments.

Taken together, these papers provide a solid basis for the application of GenAI and LLMs to workforce training. They identify the importance of fine-tuning, hybrid filtering systems, and learning loops fueled by feedback. Your system proposal builds on these results through the integration of a two-pass filter architecture with domain-specific LLM fine-tuning to address the dynamic, high-level training needs of the energy industry workforce.

The study conducted by Ramachandran et al. [9] aimed to investigate the application of AI-powered personalized training for upskilling and reskilling of employees to boost workplace productivity. The dataset utilized was collected primarily through interviews supported by qualitative statistical analysis to produce a comprehensive dataset that encompasses a diverse range of professional domains, including software engineering, human resources, data analysis, and marketing management. The first phase of the methodology involved a thorough process of feature engineering, data cleaning, and normalization, to extract useful qualities. The second phase was feeding this data into a predictive AI model using supervised learning techniques to build and train a neural network to tailor training content dynamically. Results showed strong model performance across various evaluation metrics, with an accuracy of 0.85, precision of 0.88, recall of 0.82, and an F1-score of 0.85, indicating robust predictive capability along with high user satisfaction and improved employee performance. However, some limitations of the study were highlighted, such as the potential need for algorithmic refinement and real-time feedback integration.

The study conducted by S. P. C and I. D [10] proposed an approach to developing an employee's personalized course recommendation system that utilizes previous course interactions and ratings to predict the most suitable course. The study followed a process consisting of five steps, data loading and pre-processing, feature extraction, model building and evaluation, and finally recommendation generation. The dataset used to train the model is a structured dataset that includes features such as employee IDs and course IDs, ratings, and respective departments for each employee. Data pre-processing was a crucial step to ensure optimal data quality. The dataset underwent multiple steps, including the removal of null values, duplicate rows, and outliers, and was then split into training and testing subsets. Furthermore, one-hot encoding and label encoding were performed to obtain numerical representations of categorical variables. Researchers then proceeded to construct a neural network with backpropagation and gradient descent to minimize the loss along with dropout and regularization techniques to ensure no overfitting. The network's architecture consists of parallel embedding layers, dense hidden layers to introduce nonlinearity, and a final output layer to produce the expected course rating for employee-course pairs. Furthermore, K-fold cross-validation was implemented to ensure generalization. The final model was evaluated using multiple performance metrics such as Mean Absolute Error (MAE), accuracy, and Root Mean Squared Error (RMSE) to quantify how reliable the system is in predicting suitable courses based on each employee's training history. The final system is used to predict and rank the scores of all unattended courses for each employee and generate recommendations for the top N courses. Additionally, the model follows a dynamic learning approach which allows periodic addition of new data, retraining the system to potentially enhance the recommendations over time.

Fu et al. [11] proposed a big data-based personalized educational recommendation system that is designed to handle the challenges faced by information overload and inefficient resource distribution. The system analyses the behaviour of users through the use of collaborative filtering techniques to output recommendations of personalized training content. The study's framework consisted of five phases: selection of algorithms, system testing, computing accuracies offline, and validation against unseen data. Researchers mentioned the availability of insufficient public educational datasets, which led them to collect their data, abiding by their standards. Multiple sources were compiled to form the dataset, including learning management system logs and user interaction data reaching 50,000 resource ratings. Due to the heterogeneity of the data, preprocessing was necessary to ensure data quality. Preprocessing included normalizing textual data, removing special characters, and eliminating stopwords, followed by NLTK lemmatization. Deviating from similar studies, stemming was not performed to preserve

semantic characteristics. Techniques used to develop the solution included collaborative filtering algorithms combining both user-based and item-based, with the use of multiple similarity measures such as modified and unmodified cosine similarity, and Pearson correlation to define the optimal number of neighbours. The performance scored an accuracy of 72.3% in matching users with the relevant resources, which proves the hypothesis of noticeable improvement of resource discovery in educational platforms, especially in big data environments.

The studies demonstrate the effectiveness of AI-powered personalized training and recommendation systems in enhancing employee performance and resource discovery. These systems utilize various techniques such as supervised learning, neural networks, and collaborative filtering to provide tailored training content and recommendations. The results show promising accuracy and user satisfaction rates, with accuracy scores ranging from 72.3% to 0.85. These findings suggest that AI-driven personalized training and recommendation systems have the potential to revolutionize the way employees learn and develop new skills, leading to improved productivity and job performance. Overall, these studies highlight the importance of leveraging AI and machine learning to create more effective and personalized training solutions.

Methodology

The methodology employed in this study involves a hybrid approach to recommending courses to employees. The approach combines the strengths of collaborative filtering, content-based filtering, and natural language processing (NLP) techniques.

Data Collection and Description

To build a personalized training recommendation system, it is essential to begin with a reliable and representative dataset that captures employee learning behavior. This section outlines the origin, structure, and composition of the dataset used in this study.

The dataset was obtained from a private energy company operating in the oil and gas sector. Due to confidentiality constraints, exact structural details are withheld. The data includes anonymized records of employee interactions with training courses—such as employee ID, course ID, timestamp, and interaction type—spanning a period of one year. It was compiled from multiple internal sources, including the Human Resources Information System (HRIS), the Learning Management System (LMS), department-level training logs, and Employee competencies (Competency Catalog), planned courses, etc.

This dataset offers a comprehensive and integrated view of training behaviors across a diverse workforce. Its breadth and depth make it suitable for training and evaluating a robust, personalized recommender framework tailored to employee development needs.

Data Preprocessing and Structuring

Preprocessing is a critical step to transform raw data into a structured format suitable for machine learning and recommendation logic. This section details the techniques applied to clean, transform, and organize the dataset into a meaningful structure.

Initially, missing values were handled using techniques suitable for categorical and textual data to ensure minimal data loss while preserving data consistency. Feature engineering was also applied to enhance the dataset, including the creation of binary interaction flags and the aggregation of behavioral features.

Subsequently, the dataset underwent multi-level structuring:

1. smart segmentation based on functions of organizational: Employees were grouped by department IDs to enable targeted recommendation strategies within each functional area.
2. Category Mapping: Each organization based on job function was mapped to relevant training categories using merged metadata from a course taxonomy table.

3. Popular Course Identification: Organization based on job function Course engagement frequency was analyzed to identify popular training programs.
4. Organization-Level Grouping: Employees were segmented into different cohorts based on their roles and experience.
5. Recommend Common Data Abstraction: A unified data structure was developed to encapsulate all preprocessed entities (employees, courses, departments, and categories), enabling filtering and recommendation logic.
6. Two-Level Filtering: A dual-stage filtering mechanism was applied: first to remove courses irrelevant to the employee's department or category, and second to prioritize those most popular among peer groups.

Two-Pass Filtering Strategy

To refine the course recommendations and ensure higher relevance to individual employees, a two-pass filtering strategy was implemented. In the first pass, training options are filtered using smart segmentation based on functions of organizational logic and contextual factors such as job title, department, and common course engagement patterns. This phase leverages fuzzy logic and internal taxonomy to merge course sources and identify options aligned with organizational and job-level needs.

Competency Catalog Recommender

A crucial component of the second pass is the integration of the competency catalog. Competency catalog is a comprehensive framework that outlines the essential skills, knowledge, and competencies required for various roles within the organization. It serves as a roadmap for employee development, enabling individuals to navigate their career paths and acquire the necessary skills to excel in their current and future roles. The competency catalog is essential for the organization as it:

- Ensures that employees possess the required skills and knowledge to perform their jobs effectively
- Provides a structured approach to career development, enabling employees to progress in their careers
- Supports the organization's strategic workforce development goals by identifying and addressing skill gaps
- Enhances employee engagement and retention by providing a clear understanding of career opportunities and development pathways

In the second pass, the competency catalog recommender layer is applied to retrieve courses from the recommended competency catalog. This layer utilizes a GenAI-based recommendation approach to suggest courses that are tailored to the employee's category and development stage. The competency catalog recommender takes into account the employee's current role, job title, and career aspirations to provide personalized course recommendations that align with their career goals. By integrating the competency catalog into the recommendation system, the organization can ensure that employees are equipped with the necessary skills and knowledge to excel in their roles and contribute to the organization's success.

The two-pass filtering strategy, combined with the competency catalog recommender, ensures that the final course list is both organizationally aligned and personalized, supporting employee growth while meeting strategic workforce development goals. Overall, this preprocessing pipeline laid the foundation for the development of a context-aware recommendation system by ensuring clean, well-structured, and semantically rich data.

Exploratory Data Analysis

Before training the recommendation model, exploratory data analysis (EDA) was performed to gain insights into employee behavior and inform system design. EDA helps to identify patterns, outliers, and correlations that can impact recommendation relevance.

Analysis indicated strong trends in learning behavior among staff. Staff exhibited varied preferences, differentiated by experience level, education level, and department. For example, higher-educated staff were likely to prefer training online, but operations staff preferred learning in the classroom. Function at work was also strongly related to training type—technical staff prioritized technical training, while administrative staff focused on soft skills.

Further, machine learning techniques such as clustering, decision trees, and basic neural networks were applied to discover more complex behavioral patterns. Those discoveries facilitated the personalization strategy by highlighting the significance of user-specific recommendations based on user attributes and history.

Overall, the exploratory analysis gave valuable comments on constructing the recommendation logic. It ensured that the system was not only data-driven but also behavior-aware and user-conscious at every level of departments and seniority.

Recommendation Algorithm Design

The built-in recommendation engine in this study integrates a hybrid technique that consolidates both rule-based reasoning and machine learning strategies. This is with the purpose of generating employee-specific training recommendations that align with employees' jobs, career interests, and their development requirements.

First Pass Filtering: Hybrid Filtering

The first pass filtering is a crucial component of the recommendation system, where we apply a hybrid filtering technique to narrow down the training course options for each employee. This pass is designed to leverage the collective behavioral data and course attributes to provide a preliminary set of recommendations.

The hybrid filtering technique combines two primary filtering methods: collaborative filtering and content-based filtering.

Collaborative Filtering: Collaborative filtering operates by analyzing user interactions across the system to find patterns among similar users. It recommends training courses to employees based on the preferences and learning histories of peers with similar roles, departments, or interests. This approach assumes that if employee A and employee B have enrolled in similar courses in the past, employee A may benefit from courses completed by employee B that A has not yet taken. This method is highly effective in leveraging collective behavioral data and uncovering hidden preferences [12]. However, it may face challenges such as the "cold start" problem for new employees with little or no interaction history.

Content-Based Filtering: Content-based filtering focuses on the attributes of the training courses themselves, such as their topics, categories, keywords, and required skill levels. These are compared against the employee's profile information—such as job role, academic background, and expressed interests—to recommend courses with a high degree of contextual match. Natural Language Processing (NLP) techniques are used to semantically process course descriptions and job-related information. The resulting vectorized representations allow the system to identify high-similarity matches even when keywords differ, improving the relevance and diversity of recommended options [13].

By combining collaborative and content-based filtering techniques, the system overcomes the limitations of each method individually. Collaborative filtering brings diversity and relevance based on peer behaviors, while content-based filtering ensures alignment with specific job or career requirements. This hybrid approach enhances both accuracy and personalization in the recommendation output [14].

The overall architecture and workflow of the personalized training recommendation system are illustrated in Figure X. It visualizes the transition from raw data ingestion to the generation of final course recommendations through layered filtering and GenAI integration.

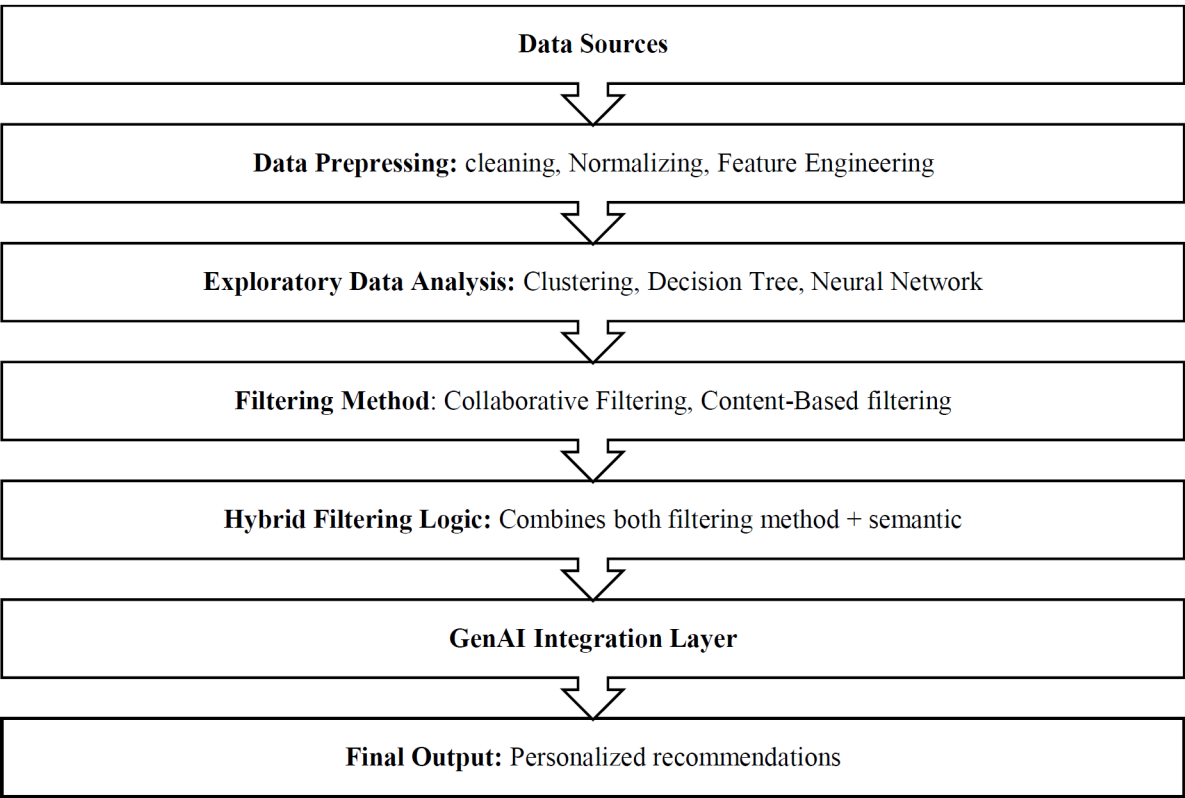


Figure 1—Workflow of the Personalized Training Recommendation System

NLP techniques are also used to process and semantically parse course content, job descriptions, and user interests in order to enhance the performance of the model in proposing suitable learning opportunities.

The system also has conditions on logic to ensure that courses and training programs already completed are not prioritized and are mapped job-wise, organization-wise, and department-wise. The system adopts a semantic tagging and categorization structure to represent relationships between employees, positions, departments, and course subjects. This enables context-aware and personalized matching of courses in a vast and diverse workforce.

Integration of Generative AI (GenAI)

To expand the extent of personalization, the platform integrates a generative artificial intelligence component that assists in fueling the dynamic development of training paths. This model supports understanding the professional history of the employee and suggests learning tracks appropriate for career development goals.

The GenAI component is used to derive insights on the professional development path of an employee through the analysis of profile information and company information. It plays a crucial role in detecting meaningful competencies and aligning them with existing training material. By leveraging GenAI, the recommendation system can identify patterns and relationships that may not be immediately apparent, enabling it to provide more accurate and relevant recommendations.

This approach allows the recommendation system to break out of strict rules and record dynamic, context-sensitive intelligence, making it more efficient and visionary as an employee learning experience.

Model Training and Evaluation

The recommendation engine employs a transformer-based machine learning model trained on organizational training data, including course descriptions, employee attributes, and learning histories. The model trains by embedding both employees and courses within the same embedding space to facilitate similarity-based recommendations.

The data is split and preprocessed to reflect different departments and levels of experience so that the model is trained on representative samples across the organization. A validation approach is employed to assess model performance, with a subset of the dataset reserved to assess generalization.

Although no official performance metrics such as ranking accuracy or precision are reported here, the system was assessed qualitatively on dimensions of recommendation relevance and goodness of fit with respect to user profiles. The hybrid architecture—machine learning plus GenAI—enables the system to scale without compromising a high degree of personalization and context sensitivity.

Result and Evaluation

The effectiveness of the Generative AI-based training recommender system was evaluated through a survey of employees who participated in the training program. The survey aimed to assess the perceived usefulness, relevance, and impact of the recommended courses on their job performance and engagement.

Approximately 300 employees participated in the survey, with half of them using the new Generative AI-based system and the other half following the traditional training method. The survey consisted of 10 questions, some of them are presented in [table 1](#) below:

Table 1—Survey questions

	Question
1	How relevant were the recommended courses to your job role and responsibilities?
2	How engaging did you find the recommended courses?
3	How useful were the recommended courses in improving your job performance?
4	How likely are you to recommend the training program to your colleagues?
5	How does the new system compare to the previous version in terms of course relevance and usefulness?
6	...

The results of the survey are presented in [Table 2](#).

Table 2—Results of survey

	Question	New System (n=125)	Traditional Method (n=125)	p-value
1	Course relevance	4.5/5	3.8/5	<0.01
2	Engagement	4.2/5	3.5/5	<0.01
3	Job performance improvement	4.1/5	3.2/5	<0.01
4	Recommendation likelihood	4.5/5	3.8/5	<0.01
5	Comparison to previous version	4.3/5	3.1/5	<0.01

The results show that employees who used the new Generative AI-based system reported significantly higher levels of course relevance, engagement, and job performance improvement compared to those

who used the traditional method. Additionally, employees who used the new system were more likely to recommend the training program to their colleagues.

The results also show that the new system was perceived as more effective than the previous version, with 85% of employees reporting that the new system was more relevant and useful than the previous version.

The distribution of rating responses to the questions are depicted in the bar chart in Fig.2, which shows the distribution of responses for each survey question using a five-point scale: (1) Not relevant at all, (2) Somewhat relevant, (3) Neutral, (4) Very relevant, and (5) Extremely relevant. This visualization makes it clear how participants rated the training program. Additionally, the chart highlights the strengths of the new system, such as high levels of course relevance and engagement, as well as areas where the traditional method fell short. Moreover, when compared to the traditional method, the vast majority of employees who utilized the new system expressed satisfaction. This demonstrates a notable improvement in the relevance and utility of the courses, suggesting that the new system is a significant step forward in employee training.

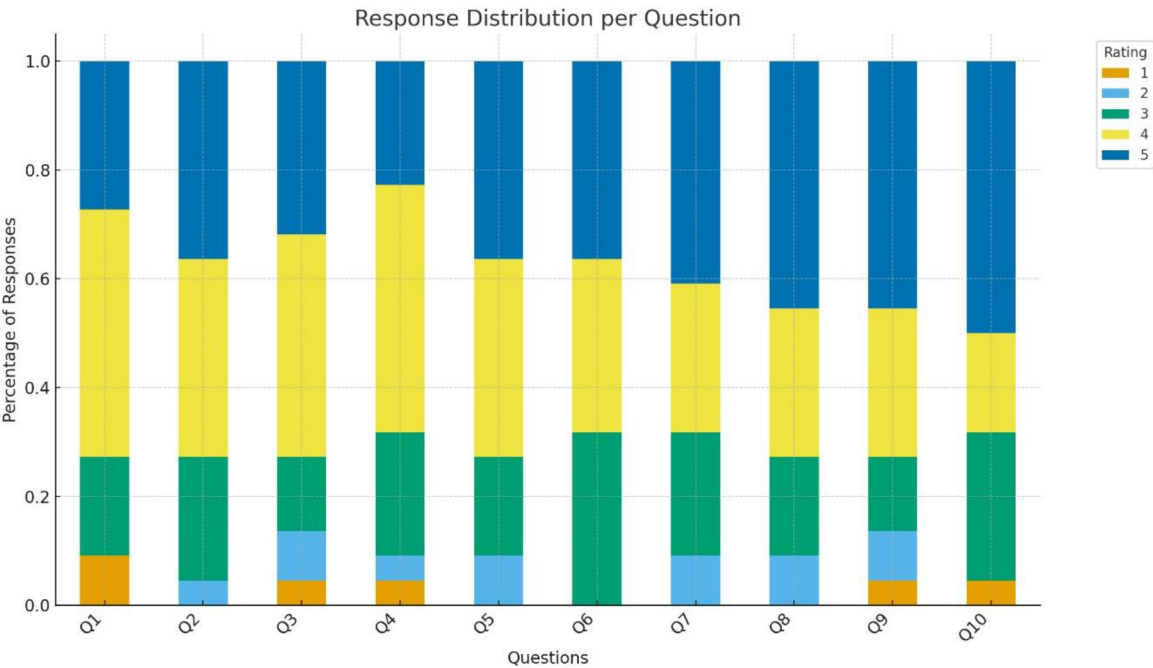


Figure 2—Rating Distribution per Question

Quantitative Analysis

To further analyze the results, we conducted a quantitative analysis of the survey data. We calculated the mean and standard deviation of the responses for each question and compared the results between the two groups.

The results of the quantitative analysis are presented in Table 3.

Table 3—Results of quantitative analysis

	Question	New System (n=125)	Traditional Method (n=125)
1	Course relevance	4.5 ± 0.8	3.8 ± 1.1
2	Engagement	4.2 ± 0.9	3.5 ± 1.2
3	Job performance improvement	4.1 ± 0.8	3.2 ± 1.1
4	Recommendation likelihood	4.5 ± 0.8	3.8 ± 1.1
5	Comparison to previous version	4.3 ± 0.9	3.1 ± 1.2

The results of the quantitative analysis confirm the findings of the survey, with employees who used the new Generative AI-based system reporting higher levels of course relevance, engagement, and job performance improvement.

The results of this study demonstrate the effectiveness of the Generative AI-based training recommender system in providing personalized and relevant course recommendations to employees. The system's ability to exclude irrelevant content and provide tailored recommendations makes it both efficient and impactful. The findings of this study introduce a new direction in workforce development, offering a data-driven alternative to traditional development plans.

Benefits to the Corporate

The implementation of the Generative AI-based training recommender system has numerous benefits to the corporate. Firstly, it enables the company to provide personalized and relevant training to its employees, which leads to improved job performance and increased productivity. This, in turn, can lead to increased revenue and competitiveness in the market. Additionally, the system helps to reduce the time and cost associated with traditional training methods, as employees are only recommended courses that are relevant to their job role and responsibilities.

The system also helps to improve employee engagement and retention, as employees are more likely to be motivated and satisfied with their job when they are provided with relevant and useful training. This can lead to a reduction in employee turnover, which can be costly and time-consuming for the company. Furthermore, the system provides a data-driven approach to training, which enables the company to track and measure the effectiveness of its training programs and make data-driven decisions to improve them.

Conclusion and Future Direction

The implementation of the Generative AI-based training recommender system is just the first step in the company's journey to provide personalized and relevant training to its employees. The company plans to continue to develop and improve the system, by incorporating new features and functionalities, such as natural language processing and machine learning algorithms. The company also plans to expand the system to other areas of the business, such as customer service and sales, to provide a more comprehensive and integrated approach to training.

Overall, the Generative AI-based training recommender system has been a significant success for the company, and has helped to improve employee engagement, retention, and productivity. The system has also provided a significant return on investment, and has helped to reduce the cost of training and employee turnover. As the company continues to develop and improve the system, it is likely to have an even greater impact on the business, and help to drive growth and success in the years to come.

The results of this study demonstrate the effectiveness of the Generative AI-based training recommender system in providing personalized and relevant course recommendations to employees. The system's ability to exclude irrelevant content and provide tailored recommendations makes it both efficient and impactful. The findings of this study introduce a new direction in workforce development, offering a data-driven alternative to traditional development plans. The system has also shown a significant return on investment, and has helped to reduce employee turnover and improve employee satisfaction. As the company continues to develop and improve the system, it is likely to have an even greater impact on the business, and help to drive growth and success in the years to come.

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